



each individual device.

- The goal is to make it easy for developers and sensor manufacturers to map their sensors and actuators on top of compatible hardware, and to allow high level languages and structures to govern low level communication protocols.

- Reference: <https://github.com/eclipse/mraa/tree/master/examples/python>

IoT communication protocols.

sockets. Install using ‘pip install sockets’.

- TCP/IP and UDP are communication transport layer protocols.

- sockets package facilitates TCP/IP and UDP communication, which are used to enable networking in IoT devices.

- Reference: <https://pypi.org/project/sockets/>

paho-mqtt. Install using ‘pip install paho-mqtt’.

- The MQTT or Message Queue Telemetry Transport protocol is a publish/subscribe messaging transport that is intended for machine-to-machine (M2M)/Internet of Things connectivity, enabling high-speed data exchange with low payload.

- When a minimal code footprint is necessary or bandwidth on the network is at a premium, it is beneficial for communications with faraway locations.

- Applications can connect to a MQTT broker to publish messages, subscribe to topics, and receive published messages thanks to the client class provided by the Paho library code.

- Additionally, it offers certain assistance functions to make it simple to publish one-time messages to MQTT servers.

- Reference: <https://pypi.org/project/paho-mqtt/smtplib>.

- This protocol handles the sending and routing of emails between mail servers.

- It is part of Python’s standard library.

- Reference: <https://docs.python.org/3/library/smtplib.html>

asyncio. Install with ‘pip install asyncio’.

- The asyncio module provides architecture for building single-threaded concurrent programming with co-routines, multiplexing I/O access through sockets and other resources, running network clients and servers, and other primitives.

- Reference: <https://pypi.org/project/asyncio/>

IoT database.

MySQLdb. Install with ‘pip install MySQL-python’.

- MySQLdb is a popular go-to relational database that aids in the building of remote storage for IoT systems.

- Reference: <https://pypi.org/project/MySQL-python/SQLite>.

- SQLite is a C library that provides a lightweight disk-based database that does not require a separate server process and may be accessed using a non-standard variation of SQL.

- SQLite can be used by IoT to store internal data.

- The sqlite3 module does not require installation. Since Python 2.5, it has been included in the standard library.

- Reference: <https://docs.python.org/3/library/sqlite3.html>

IoT graphical user interface.

Tkinter.

- Python provides numerous GUI (graphical user interface) development choices. Tkinter is the most commonly used of all the GUI techniques.

- It is a standard Python interface to the Python-supplied Tk GUI toolkit.

- Python with Tkinter is the quickest and most straightforward approach to construct GUI apps.

- Reference: <https://docs.python.org/3/library/tkinter.html>

IoT backend.

Flask. Install via ‘pip install Flask’.

- Flask is a simple micro framework for building web applications, to build http requests.

- IoT applications require a web-based interface in order to provide monitoring and controlling devices.

- Reference: <https://pypi.org/project/Flask/>

websockets. Install using ‘pip install websockets’.

- websockets is a Python package for creating WebSocket servers and clients. It prioritises consistency, simplicity, resilience, and performance.

- The WebSocket API is a cutting-edge technology that allows a two-way interactive communication session to be established between a user’s browser and a server. This API sends messages to a server and receives event-driven responses without polling the server for a response.

- Reference: <https://pypi.org/project/websockets/requests>. Install with ‘pip install requests’.

- The requests module allows sending HTTP requests; the HTTP request generates a ‘response object’ containing response data (content, encoding, status, and so on).

- Supports multipart file uploads, streaming downloads, and connection pooling.

- Enables automatic decoding and decompression of content.

- Reference: <https://pypi.org/project/requests/>

aiohttp. Install with ‘pip install aiohttp’.

- aiohttp provides an asynchronous HTTP client or server framework, including websockets support.

- Reference: <https://pypi.org/project/aiohttp/>